Sai Yashwanth Dasari

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Summary

Full-stack engineer (B.Tech, AI/ML, 2025) who builds end-to-end products with Python and React: REST APIs, data models, authentication and background jobs, plus LLM work with RAG and multi-agent pipelines. Research intern at the University of Hyderabad (GAN-based Telugu song pattern recognition, paper submitted to IHCI 2023). Delivered 15+ Django REST APIs in a five-month internship; built and deployed CentSible (expense tracker) and ClearHaul (truck routing). Comfortable with AI coding assistants. 4 merged pull requests to a community FastAPI starter template; 398 LeetCode problems solved.

Education

J.B. Institute of Engineering and Technology — Hyderabad, Telangana
Bachelor of Technology in Artificial Intelligence and Machine Learning | CGPA: 7.88/10.0 — *2021 – 2025*

Experience

Software Engineer Intern — Dec 2024 – April 2025
Workcohol — *Chennai, Tamil Nadu*

- Built **15+ REST APIs** in **Django REST Framework** for a survey platform, in a team of 4 developers
- Optimized **MySQL** queries for **40% faster** response times; designed an indexed schema for 10+ question types with complex relationships
- Implemented **JWT authentication** and role-based access control across **3 user roles**

Deep Learning Research Intern — Sep 2023 – Jan 2024
University of Hyderabad — *Hyderabad, Telangana*

- Researched **GAN-based Telugu song pattern recognition** and wrote a short paper on it, submitted to **IHCI 2023**
- Extracted audio features with **Librosa** and **Praat** and processed them with Pandas and NumPy
- Built **SVM** and **GAN** models in **TensorFlow** for classification tasks

Projects

CentSible: Personal Expense Tracker — Oct 2026 – Present
FastAPI, React, Material UI, PostgreSQL, Redis, Celery, RabbitMQ, Docker — GitHub | Live Demo

- Built and deployed a full-stack expense manager with **30+ REST endpoints**: **FastAPI** and async **SQLAlchemy 2.0** on **PostgreSQL** (**Supabase**) with **Alembic** migrations, and a **React** and **Material UI** frontend; filtered expense CRUD, CSV export with formula-injection protection, budgets, insights and recurring expenses
- Implemented cookie-based **JWT auth** (HttpOnly access and refresh tokens, sessions invalidated on password change) with **Google OAuth 2.0**, hashed single-use password-reset tokens and **Redis** rate limiting
- Added a one-click **demo account** that creates an isolated guest user with sample data, protected by per-IP rate limiting, so recruiters can try the live app without signing up
- Offloaded emails to **Celery** workers on **RabbitMQ** with exponential backoff; a **Celery beat** job creates recurring expenses and is safe to run twice using **row-level locking**; deployed with **Docker** on **Render** and **Vercel**

ClearHaul: Weather-Aware Truck Routing — Oct 2026
Django, Django REST Framework, React, Leaflet, OSRM, Open-Meteo — GitHub | Live Demo

- Built a full-stack app that gets **3 routes** from OSRM, forecasts wind, rain and snow at each checkpoint for the **truck's own arrival time there**, applies load-weight rules, and recommends the safest route
- Kept risk rules and route ranking in a **pure-Python domain layer** tested without a network; ranks routes by **miles at each risk level**, not checkpoint counts, so results do not depend on the sampling interval
- Fetches all forecasts in **one batched, parallel Open-Meteo request**; the response holds 49 hours of risk per checkpoint, so the **0 to 48 h heatmap slider** and moving truck marker need **no extra API calls**; hosted on **Render** and **Vercel**.Backed by **88 automated tests** (**59 backend**, **29 frontend**), all passing

Insurance Claim Document Processing Pipeline April 2026
FastAPI, LangGraph, Gemini API, PyMuPDF GitHub

- Built a **multi-agent LangGraph** pipeline that classifies claim PDFs into **9 document types** with Gemini vision and routes each to a specialized extraction agent, with backoff retries, JSON fallbacks and a mock testing layer

Hotel Booking Analytics and RAG Q&A 2025
FastAPI, pandas, scikit-learn, OpenRouter LLM, Docker, Render GitHub | Swagger Demo

- Built a **RAG** question-answering API over **119K+ hotel booking records**: condensed the data into **346 summary documents** (monthly, yearly, per country, per segment), retrieved them with **TF-IDF** cosine similarity and answered with a free **LLM via OpenRouter**
- Routed common metric questions (revenue, cancellation rate, lead time) to exact **pandas** calculations and the rest to retrieval plus LLM, with a fallback when the LLM is down; **FastAPI** endpoints return charts as base64 images; covered by 14 **pytest** tests and deployed with **Docker** on **Render**

RouteWise: Route Stop Optimizer June 2026
Django, Django REST Framework, NumPy, PostgreSQL, OpenRouteService API GitHub | Live Demo

- Built a route-based stop optimizer over **8,100+ stations** in **PostgreSQL**, with one OpenRouteService call per request and a three-tier location lookup (cache, own database, live geocoder)
- Cut a long multi-state query from about **44 s to 11 s** (4x) with route downsampling and vectorized **NumPy haversine** checks; shipped a greedy strategy after prototyping a **dynamic-programming** variant

Smart Task Analyzer Nov 2025
Django, Django REST Framework, pytest, JavaScript GitHub | Live Demo

- Built a task prioritization system using the **Weighted Sum Model** with 4 strategies, dependency handling and penalty multipliers; **100% pytest coverage** on the REST API

Technical Skills

Frontend: React, Vite, Material UI, React Router, Leaflet, JavaScript
Backend & Auth: Python, Django, Django REST Framework, FastAPI, Flask, Pydantic, JWT, OAuth 2.0, RBAC
Data & Queues: PostgreSQL, MySQL, SQLAlchemy (async), Alembic, Celery, RabbitMQ, Redis, AsyncIO
Testing & DevOps: pytest, Docker, Docker Compose, NGINX, Vercel, Render, Git/GitHub, Linux
AI & Algorithms: LangGraph, LangChain, Gemini API, RAG, FAISS, TensorFlow, Dynamic Programming, MCDM